

Now, Solve Puzzles Using Linear Algebra

This seventh article in the mathematical journey through open source solves puzzles using linear algebra in Octave.



Matrix maths is formally called linear algebra. We have covered its basics in an earlier article. Now, let's apply what we learnt by solving puzzles.

Solving 'purchase puzzles'

Shrishti purchased 24 pencils and 12 erasers for ₹ 96. Divya purchased 20 pencils and 15 erasers for ₹ 100. What are the prices of the pencil and the eraser?

Assuming that 'p' is the price for pencils and 'e' is the price for erasers, we have the following two equations:

$$24 * p + 12 * e = 96$$

$$20 * p + 15 * e = 100$$

Hence, we could get the values of 'p' and 'e' by solving these equations. Converting them into linear algebra form, they can be re-written using matrix multiplication as follows:

$$\begin{bmatrix} 24 & 12 \\ 20 & 15 \end{bmatrix} \begin{bmatrix} p \\ e \end{bmatrix} = \begin{bmatrix} 96 \\ 100 \end{bmatrix}$$

...which is $Ax = b$, 'x' being the vector with variables 'p' and 'e'. Hence, we need to find the value of 'x', which is

given by: $x = A^{-1}b$. Using Octave:

```
$ octave -qf
octave:1> A = [
> 24 12
> 20 15
> ];
octave:2> b = [
> 96
> 100
> ];
octave:3> x = inv(A) * b
x =

    2.0000
    4.0000

octave:4>
```

Hence, $p = ₹ 2$ and $e = ₹ 4$, i.e., each pencil costs ₹ 2 and an eraser costs ₹ 4. You may check by substituting these